

Practice with continuous variables in R, Histograms, Box Plots, and Density Plots

1 Continuous Variables in R

In this notes, we will practice with continuous variables in R, focusing on histograms, box plots, and density plots. We will start by loading the necessary libraries and the “diamond” dataset.

```
library("tidyverse")
library("here")
library("patchwork")
library("krulRutils")

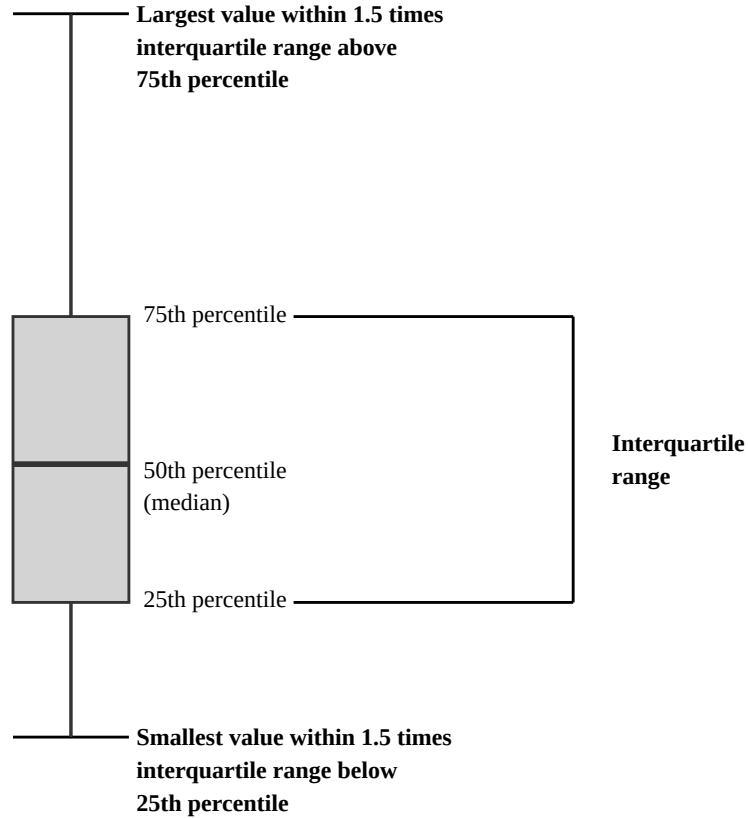
options(scipen = 999) # Disable scientific notation
data(diamonds)
```

1.1 Box Plots

Box plots are a great way to visualize the distribution of continuous variables. The following figure shows the main components of a box plot:

```
boxplot_explanation_plot <- ggplot_box_legend()
ggsave(
  filename = here("images", "boxplot_explanation.png"),
  plot = boxplot_explanation_plot,
  width = 6,
  height = 12,
  dpi = 300
)
boxplot_explanation_plot
```

500 **Number of values**



- **Outside value**
 – Value is >1.5 times and <3 times the interquartile range beyond either end of the box

Figure 1: Description of the main components of a boxplot